higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ROADS
NQF LEVEL 3

(12030363)

21 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 5 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1: FURTHER ROAD BUILDING TERMINOLOGY

1.1 Most of the roads in SA are finished off with asphalt to make the traffic flow easier.

List THREE different types of materials used to compose asphalt. (3)

1.2 The leveling course is made up by one or more layers of asphalt with a minimum thickness specified by an engineer and can be applied to the road that is being resurfaced.

State THREE functions of the leveling course. (3)

1.3 A culvert is a pipe used to direct water away from the road.

List FOUR other uses of a culvert under the carriageway. (4)

1.4 Define the following concepts:

1.4.1 Carriageway

1.4.2 Grade line

1.4.3 Haul distance

1.4.4 Road prism

1.4.5 Borrow material

(5 × 2) (10)
[20]

QUESTION 2: INTRODUCTION TO ROAD SURFACING

2.1 Every road must be surfaced because gravel roads cannot handle excessive loads in very wet weather conditions.

State THREE benefits of the surfaced road. (3)

2.2 Give FIVE steps to be followed when preparing the base for priming. (5)

2.3 List THREE types of asphalt used on the road. (3)

2.4 Why is an asphalt paver fitted with automatic electronic screed control? (1)

2.5 State THREE materials used to make a double seal. (3)
[15]

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QUESTION 3: ROAD FURNITURE

3.1 List THREE different types of road furniture found on the road side. (3)

3.2 Name FIVE specific conditions for installing a guardrail on the road. (5)

3.3 What are the TWO components of guardrails? (2)
[10]

QUESTION 4: ROAD SURVEYING

4.1 What is the difference between the *horizontal curve* and *vertical curve*? (2)

4.2 Name THREE types of horizontal curves and indicate their differences. (6)

4.3 Give THREE types of sight distance. (3)

4.4 What are the THREE main stages of setting up a theodolite? (3)

4.5 A theodolite is an instrument used to measure angles.

State SIX procedures to be taken when using a theodolite to measure the horizontal angle. (6)

4.6 Name FIVE different parts of a theodolite. (5)
[25]

QUESTION 5: EARTHWORKS

5.1 List FIVE different types of rollers used to compact the soil during road construction. (5)

5.2 Name the plant used to compact the following types of soil and state ONE reason for compacting it:

5.2.1 Salty sand and clay

5.2.2 Granular material

5.2.3 Crushed stone

(3 × 2) (6)

5.3 Give ONE characteristic of the following types of soil when it is dry:

5.3.1 Cohesive soil

5.3.2 Non-cohesive soil

(2 × 1) (2)

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5.4 List FIVE different types of materials needed when you perform an AASHTO test. (5)

5.5 What is the other name of AASHTO? (1)

5.6 What are the TWO things that the modified AASHTO do on the soil? (2)

5.7 Write the following abbreviations in full:

5.7.1 MC

5.7.2 OMC

5.7.3 RD

5.7.4 AASHTO

5.7.5 CBR

(5 × 1) (5)

5.8 What is the relationship between maximum dry density and optimum moisture content? (4)

[30]

TOTAL: 100

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